GOOGLE CLOUD SKILLS | HANDS-ON LAB REPORT

Manage Access Control with IAM

Identity and Access Management · Lab ID: GSP064

PREPARED BY	Yabesh J
DATE COMPLETED	20 October 2026
PLATFORM	Google Cloud (IAM, Cloud Storage)
LEVEL / DURATION	Introductory / 15 minutes
FINAL SCORE	100 / 100

1. Overview

Google Cloud **Identity and Access Management (IAM)** lets you create and manage permissions for Google Cloud resources. It brings access control for all Google Cloud services into a single system with a consistent set of operations.

In this lab I used two sets of credentials, Username 1 (Project Owner) and Username 2 (Project Viewer), to see how granting and revoking IAM roles changes what a user can do.

2. Objectives

- Explore IAM roles in the Google Cloud console
- Assign a role to a second user
- Remove assigned roles
- Verify access using Cloud Storage and Cloud Shell

3. Concepts and Resources

Basic roles (project-level)

Role	Permissions
roles/browser	Read access to browse the project hierarchy (folders, organizations and IAM policies). Does not include permission to view resources in the project.
roles/viewer	Read-only actions that do not change state, such as viewing existing resources or data.
roles/editor	All viewer permissions, plus actions that modify state, such as changing existing resources.
roles/owner	All editor permissions, plus managing roles and permissions for the project and setting up billing.

Resource	Details
Project	qwiklabs-gcp-04-a400e049008a
Bucket	yabeshbuckets – US multi-region, Standard class, not public
Object	sample.txt (62.9 KB), uploaded to the bucket
Predefined role used	Cloud Storage > Storage Object Viewer, granted to Username 2

The project and users were temporary and were removed when the lab ended.

4. Step-by-Step Walkthrough

Signing in as two users

I signed in to the Google Cloud console in two separate Incognito windows, one as **Username 1** and one as **Username 2**, so that I could change permissions in one window and check the effect in the other.

Task 1 – Explore IAM and project-level roles

As Username 1, I opened *IAM & Admin* > *IAM* > *Grant access* and looked at the four basic roles (Browser, Viewer, Editor, Owner) in the role picker. Username 1 has Owner permissions, so it can manage roles for the project. As Username 2, the IAM page showed the Viewer role, and the **Grant access** button was disabled, because changing IAM policy needs the *resourcemanager.projects.setIamPolicy* permission.

Task 2 – Prepare a Cloud Storage bucket

As Username 1, I created a bucket with a globally unique name (**yabeshbuckets**) in the US multi-region, uploaded a file, and renamed it to **sample.txt**. Username 2, with the Viewer role, could see the bucket.

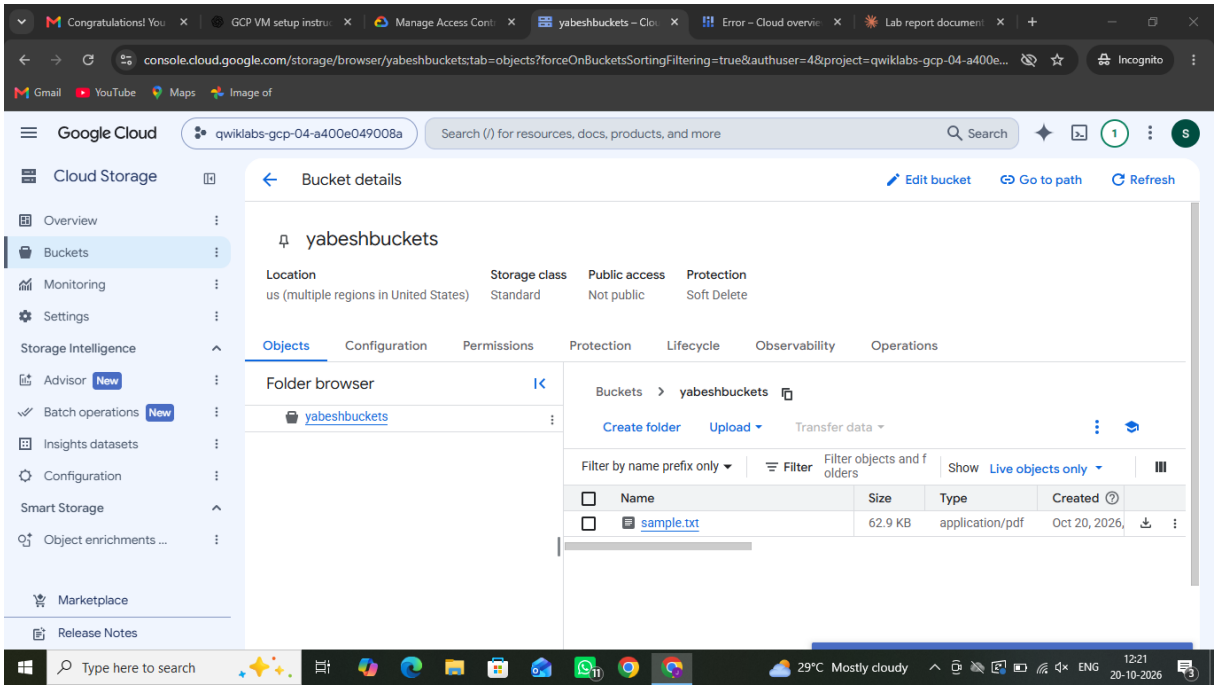


Figure 1: Bucket yabeshbuckets (US multi-region, Standard, not public) containing the uploaded sample.txt.

Task 3 – Remove project access

As Username 1, I opened the IAM table, clicked **Edit principal** (the pencil icon) for Username 2, deleted the Viewer role with the trash icon, and saved. Username 2 disappeared from the list. Back in Username 2's window, opening Cloud Storage then returned a permission error, which shows that access had been revoked. IAM changes can take up to 80 seconds to propagate.

Task 4 – Add Cloud Storage permissions

I granted Username 2 a narrower role, **Cloud Storage > Storage Object Viewer**, with *Grant access* in the IAM page, using Username 2's email as the new principal.

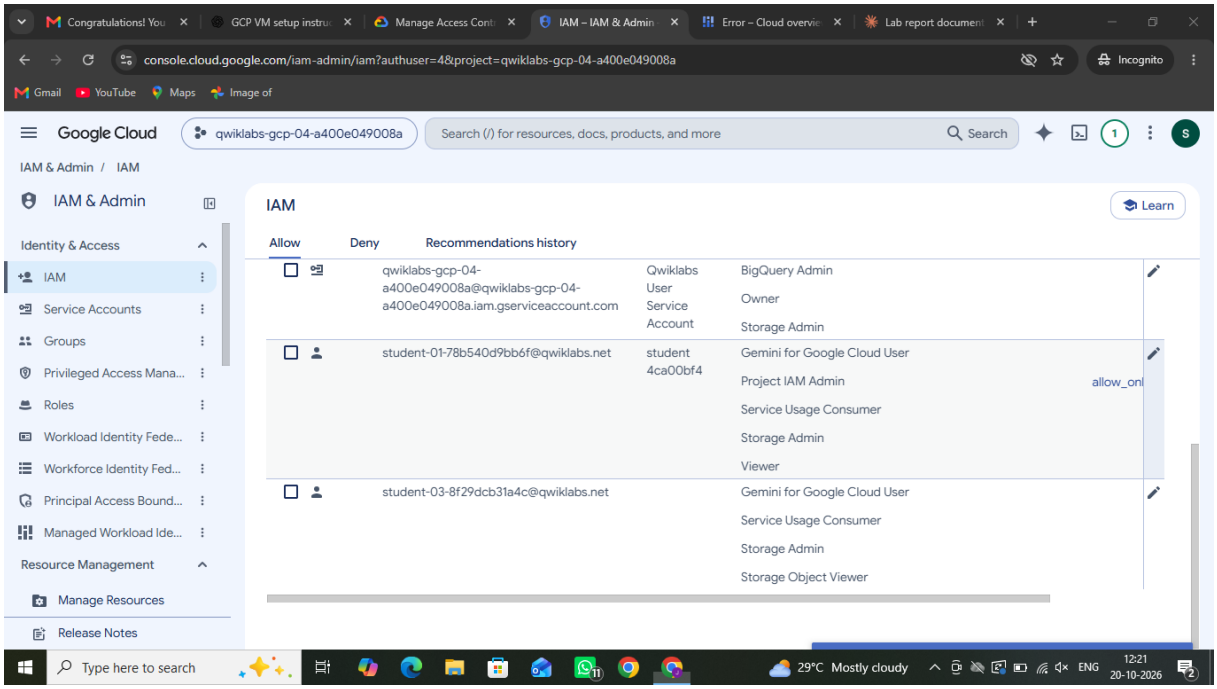


Figure 2: IAM page listing principals and their roles. Username 2's row includes Storage Object Viewer, and Username 1's row includes Project IAM Admin, Storage Admin and Viewer.

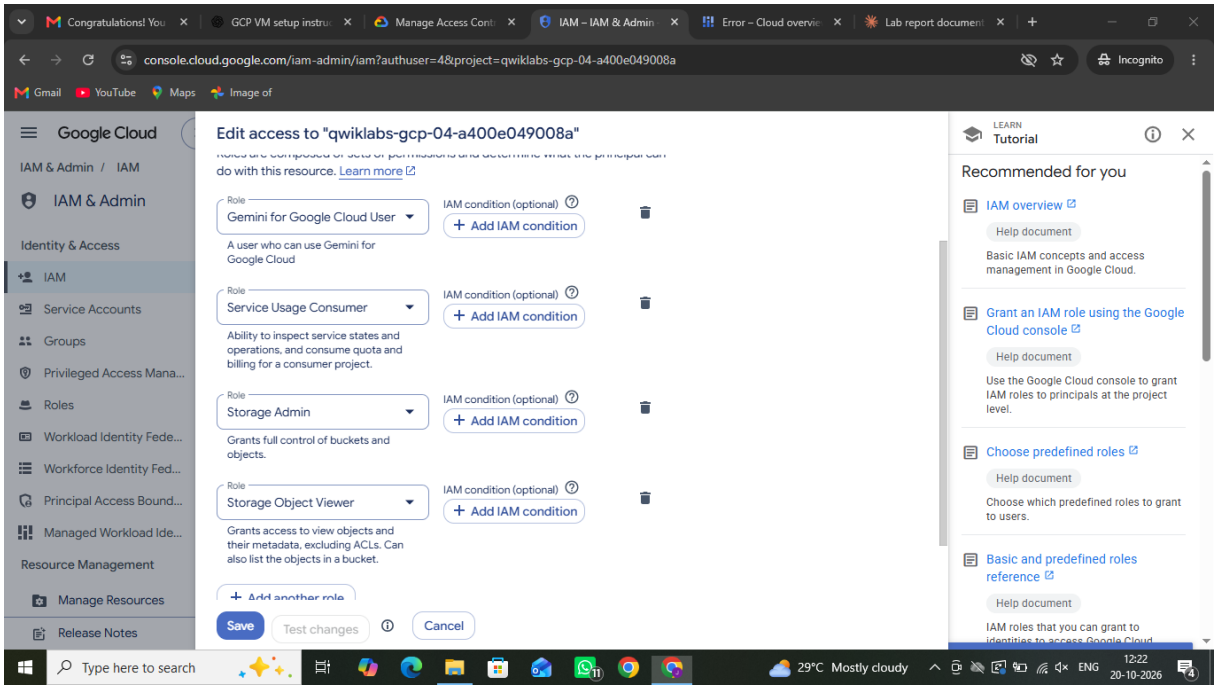


Figure 3: Edit access pane for Username 2, showing the roles assigned (Gemini for Google Cloud User, Service Usage Consumer, Storage Admin, Storage Object Viewer) with a delete icon beside each role and an "Add another role" option.

Without the Project Viewer role, Username 2 cannot browse project resources in the console, but the Storage Object Viewer role gives specific access to bucket objects. To verify, Username 2 opened Cloud Shell and listed the bucket:

```
gcloud storage ls gs://yabeshbuckets
```

The command lists `gs://yabeshbuckets/sample.txt`, which shows that Username 2 can view the objects. If an AccessDeniedException appears, waiting a minute for the permission to propagate fixes it.

5. Lab Completion

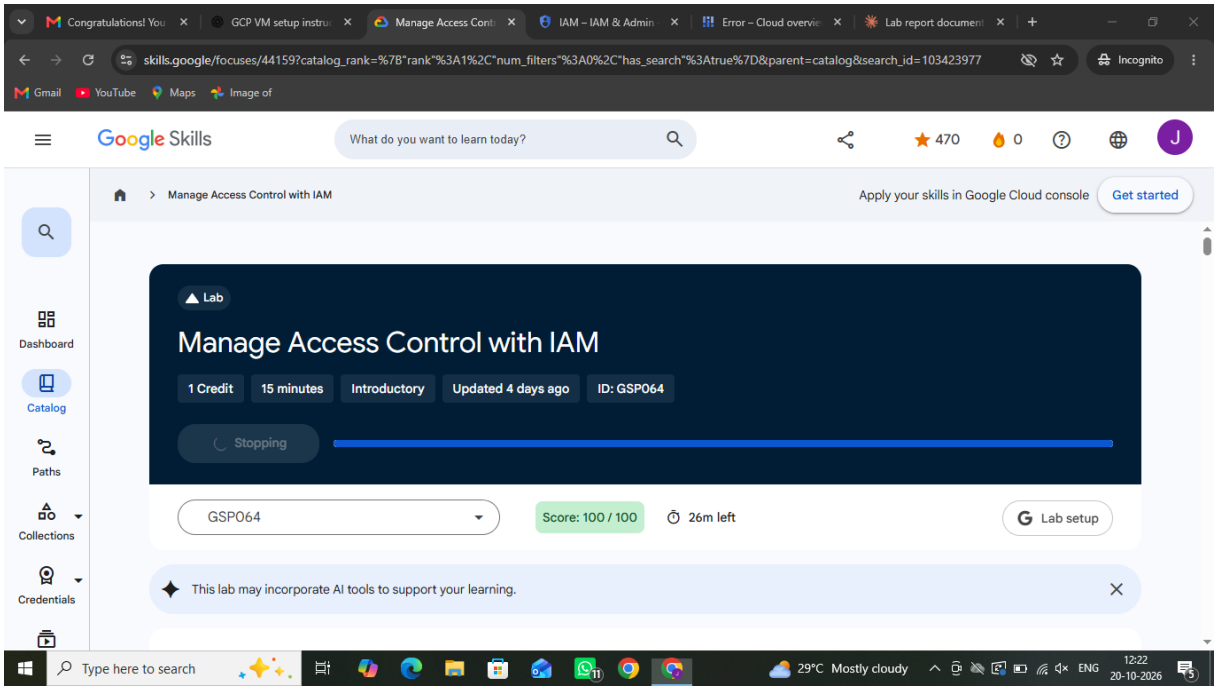


Figure 4: Lab completed with a score of 100/100.

6. Key Takeaways

- IAM controls **who** (principal) can do **what** (role) on **which resource**.
- **Basic roles** (Viewer, Editor, Owner, Browser) apply to a whole project. They are broad, so narrower predefined roles are usually better.
- The **Viewer** role gives read-only access, and it cannot change IAM policy. Granting roles needs the *setIamPolicy* permission, which Owners have.
- Removing a role takes access away. Changes can take up to about 80 seconds to apply, so wait before testing.
- A **predefined role** such as Storage Object Viewer gives access to one service without granting project-wide access. This follows the principle of least privilege.
- Testing with two identities in separate windows is a quick way to check that a policy does what you intend.

7. Conclusion

The lab was completed with a score of 100/100. I explored basic roles, created a bucket, revoked a user's Viewer role, and gave the same user the more specific Storage Object Viewer role, then confirmed the access with Cloud Storage and Cloud Shell.